# Advanced Computer Graphics (ACG)

Theory Session 1 - Introduction

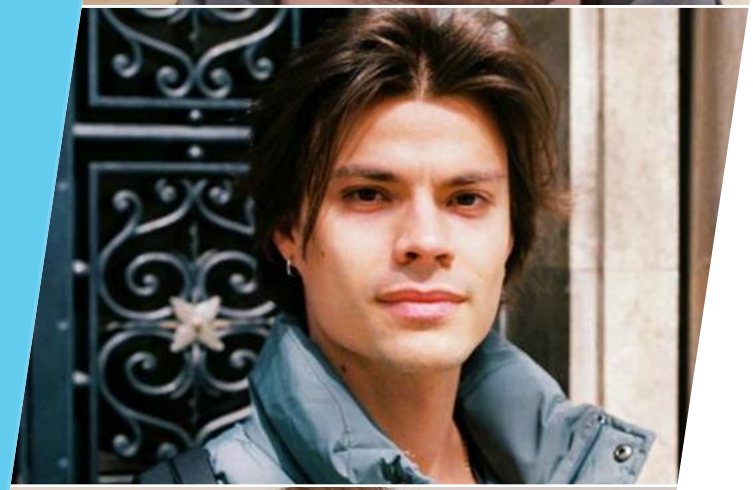
2026-2027

Pompeu Fabra University

Ricardo Marques (ricardo.marques@upf.edu)

# Course Organization







# ACG Teaching Team

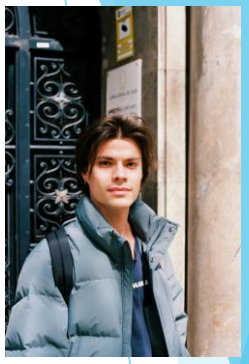
- Ricardo Marques (Theory)
  - ricardo.marques@upf.edu
- Arnau Colom (Seminars and Labs, Ray Tracing)
  - arnau.colom@upf.edu
- Víctor Ubieto (Seminars and Labs, Volume Rendering)
  - victoremilio.ubieto@upf.edu

# GTI Research Team

- GTI: Grup de Tecnologies Interactives
  - https://www.upf.edu/web/gti/demos
  - https://github.com/upf-gti
  - https://twitter.com/GTI_UPF

- Computer Graphics
- Advanced Computer Graphics
- Electronic Games
- Real-Time Graphics
- 3D Animation
- Virtual Communication Environments











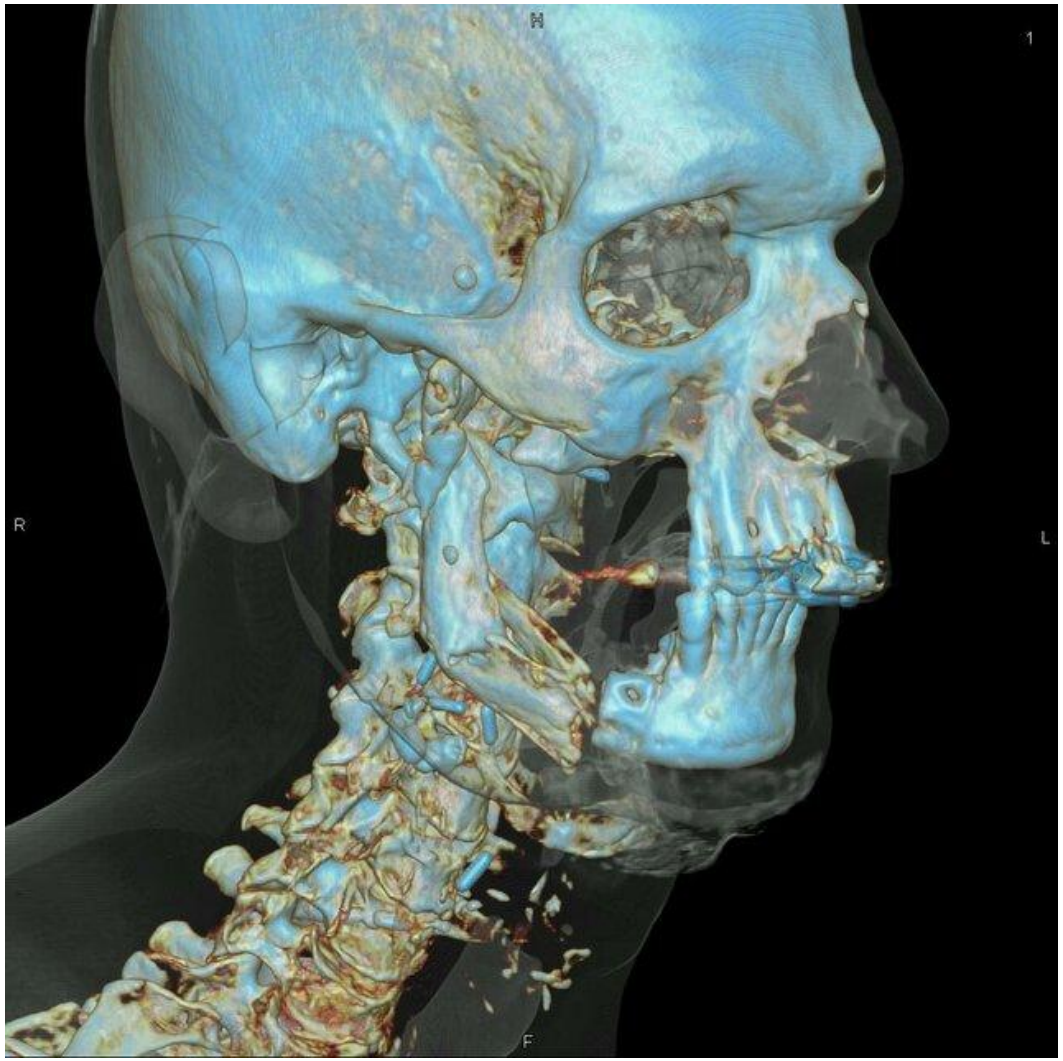

# Course Objectives

- Aspects of Computer Graphics for Cinema and Biomedicine
  - Ray Tracing
  - Volume Graphics
- C++, OpenGL and GLSL (shaders)
- A Practical Approach
  - Theory, Seminars and Labs quite intertwined
  - Expectation of **progressive autonomy**




# Theory Content

- **Block 1: Ray Tracing**
  - Introduction
  - Basic Ray Tracing
  - Radiometry, The Rendering Equation and Monte Carlo Integration
  - Direct Illumination, Perfect Reflections and Transmissions
  - Indirect Illumination and Path Tracing




# Theory Content

- **Block 2: Volume Rendering**
  - Basic Volume Rendering Concepts
  - The Volume Rendering Equation
  - Volume Rendering Approaches and Algorithms
  - Performance and Image Quality Considerations

# Lab and Seminar Content

- **Assignment 1: Ray Tracing**
  - Apply Basic and Advanced Ray-Tracing concepts to develop your own Path-Tracer
- **Assignment 2: Volume Rendering**
  - GPU-based Ray-marching for rendering volumetric data
- Details of the assignments will be given by the lab/seminar teachers

# Assessment

- **Final Grade = 1/3 Theory Grade + 2/3 Lab Grade**

- **Theory Grade:**
  - Midterm Exam (50%)
  - Final Project (50%)
    - Groups of 4 students
    - 1st presentation outline in 2-3 slides
    - 2nd presentation (2 lectures)
    - Must provide individual feedback on the other presentations

# Assessment

- **Final Grade = 1/3 Theory Grade + 2/3 Lab Grade**

- **Lab Grade (*not recoverable*):**
  - Delivery 1 on Ray-Tracing (50%)
  - Delivery 2 on Volume Rendering (50%)
  - For each delivery:
    - Report, code, interview

**Create groups (of 2)!!**

# Theory Assessment Schedule

- **Midterm exam at the end of the theoretical exposition block**
  - Week 7 (Theory Session 9, T9)
- **Presentations**
  - Each group of **4 people** (usually, two groups of labs) will work on a topic
  - Choose topic and say why
  - Week 9 – Outlines submitted and presented, receive individual feedback from other students
  - Week 10 – Full presentations + full documentation
  - Everything is marked

# Calendar

| Week no. | Date | Theory | Practice | Seminar | Deliveries |
|---|---|---|---|---|---|
| Week 1 | 28 Sep<br>02 Oct | **T1**: Introduction | **P1**: Introduction to A1.1 | - | |
| Week 2 | 05 Oct<br>09 Oct | **T2**: Basic Ray Tracing | | **S1**: Work on A1 | |
| Week 3 | 12 Oct<br>16 Oct | **T3**: Ray Casting, Recursive Ray Tracing | **P2**: Introduction to A1.2 | | |
| Week 4 | 19 Oct<br>23 Oct | **T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination<br>**T5**: Indirect Illumination and Path Tracing | | **S2**: Work on A1 | **D.A1** |
| Week 5 | 26 Oct<br>30 Oct | **T6**: Volume Rendering: Absorption-Only Model | **P3**: Introduction to A2.1 | | |
| Week 6 | 02 Nov<br>06 Nov | **T7**: Volume Rendering: Emission, Absorption and Scattering Model<br>**T8:** Neural Radiance Fields and Topics' Presentation | | **S3**: Work on A2 | |
| Week 7 | 09 Nov<br>13 Nov | **T9**: **Midterm** | **P4**: Introduction to A2.2 | | |
| Week 8 | 16 Nov<br>20 Nov | | **P5**: Work on A2 | **S4**: Work on A2 | **D.A2** |
| Week 9 | 23 Nov<br>27 Nov | **T10**: **1st Presentation** | **P6**: Work on FP | - | |
| Week 10 | 30 Nov<br>04 Dec | **T11**: **2nd Presentation** **(First Session)**<br>**T12**: **2nd Presentation** **(Second Session)** | | **S5**: Work on FP | **D.FP** |

# Course Bibliography

- **Main References:**
  - Philip Dutré, Kavita Bala, Philippe Bekaert: *Advanced Global Illumination*, A K Peters/CRC Press 2006
  - Klaus Engel, Markus Hadwiger, Joe Kniss, Christof Rezk-Salama, Daniel Weiskopf: *Real-Time Volume Graphics*, A K Peters, 2006.
- **Complementary References:**
  - Tomas Akenine-Möller, Eric Haines, Naty Hoffman: *Real-Time Rendering* (4th Edition), A K Peters/CRC Press, 2018.
  - Matt Pharr, Wenzel Jakob, Greg Humphreys: *Physically Based Rendering: From Theory to Implementation* (4th Edition), Morgan Kaufmann, 2022.
- **Secondary reference:**
  - Bernhard Preim, Charl Botha: *Visual Computing for Medicine - Theory, Algorithms, and Applications* (2nd Edition), Morgan Kaufmann, 2013.

# Plagiarism – Warning and Advice

- **Plagiarism** is the practice of appropriating the words or production of another as one's own. It is considered malpractice and it is both illegal and unethical
- **Examples**: identity theft; use of forbidden material, including apps and devices, during tests/exams/academic works; assisting someone in copying tests/exams/academic works
- The School has a commission aimed at dealing with issues related to plagiarism
- **Types of offences**: minor, serious, very serious (applied to TFGs and TFMs)
  - Specific sanctions apply in each case: warning, file open (recording the offence), fail, including loss of right to the retake in July, etc.
- **Guidelines**:
  - https://guiesbibtic.upf.edu/treball-academic
  - https://guiesbibtic.upf.edu/dretsautor/plagi
- **How to avoid it**:
  - https://www.youtube.com/watch?v=PzZsButRaHs

Universitat Pompeu Fabra Barcelona

# Soft Introduction

Introduction to Ray-Tracing

Introduction to Volume Graphics

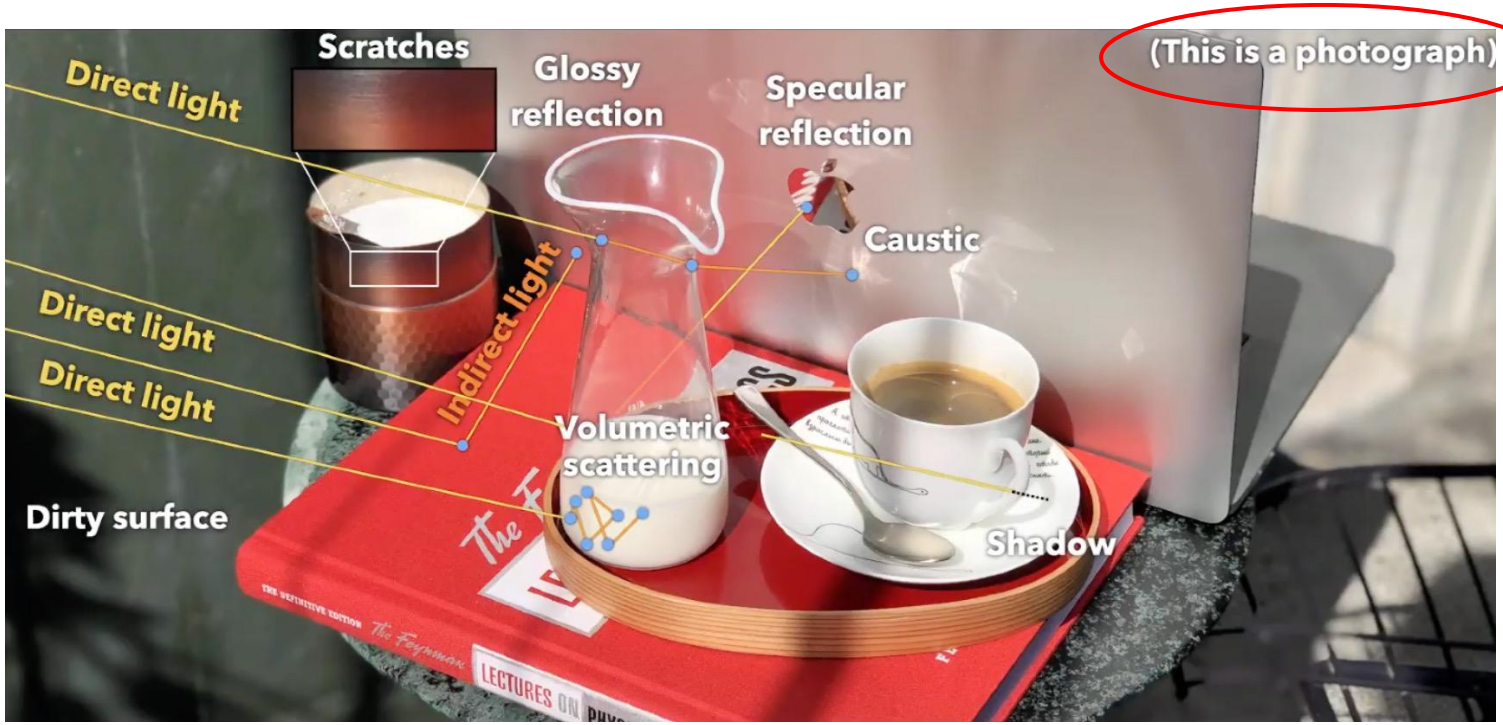


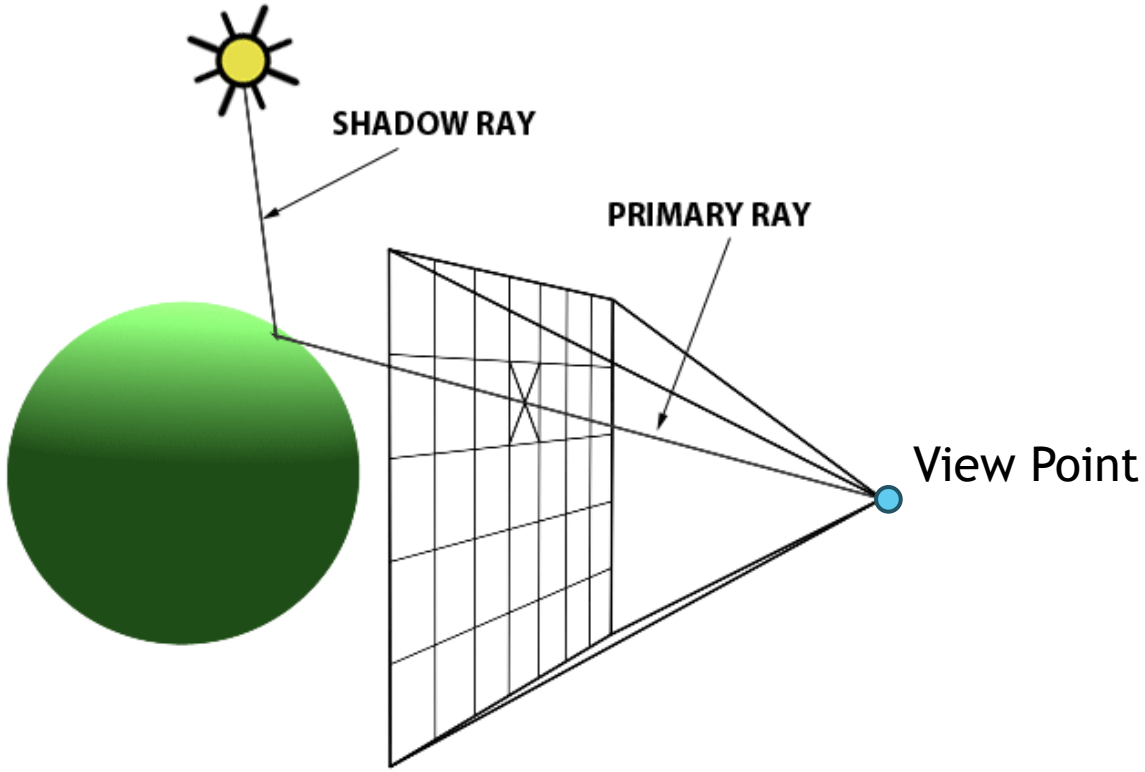

Image from a recent presentation on Physically Based Rendering (by Wenzel Jakob)

# General Course Goal: Photo-Realistic Image Synthesis

# Photo-Realistic Image Synthesis

- Photo-Realistic Image Synthesis
  - Sounds great! But how can we do so?
- Our answer: Physically-Based Rendering (PBR)
  - Set of rendering techniques
  - Simulate physical behavior of light
  - Lead to highly realistic synthetic images… and to huge rendering times
- Main tool
  - Ray-Tracing

# Ray Tracing: Basic Principle





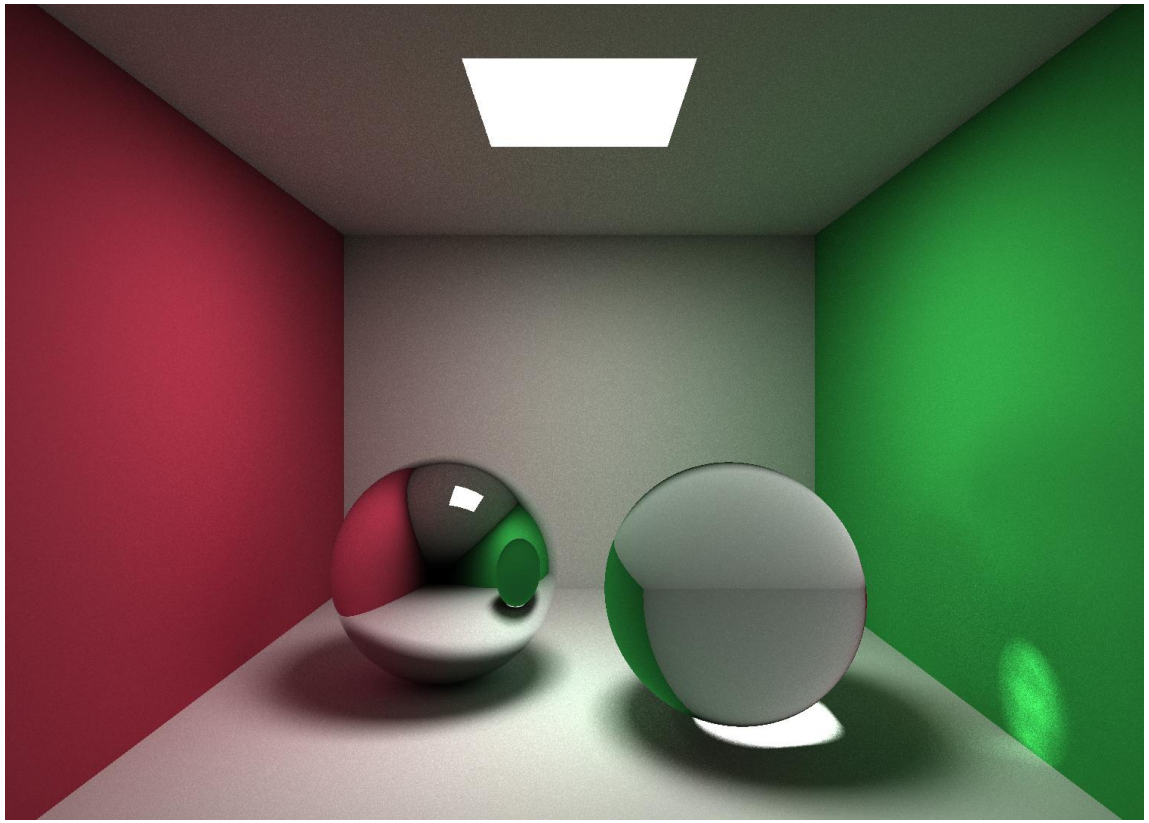


# Ray Tracing Applications



# Ray Tracing Applications

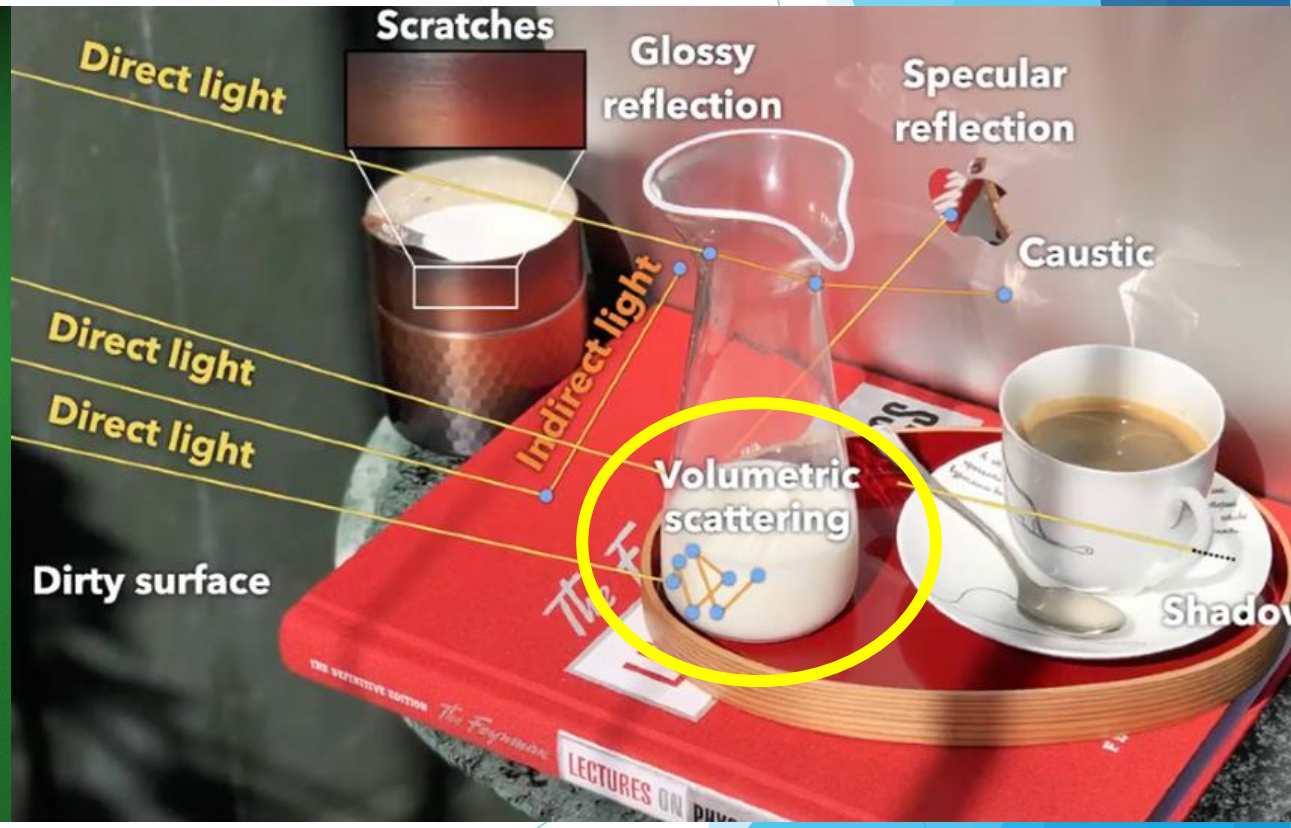
# Ray Tracing in ACG



- Example result from the Ray Tracing assignment
- Includes:
  - Direct illumination
  - Soft Shadows
  - Indirect illumination
  - Color bleeding
  - Caustics
  - Reflections and Transmissions

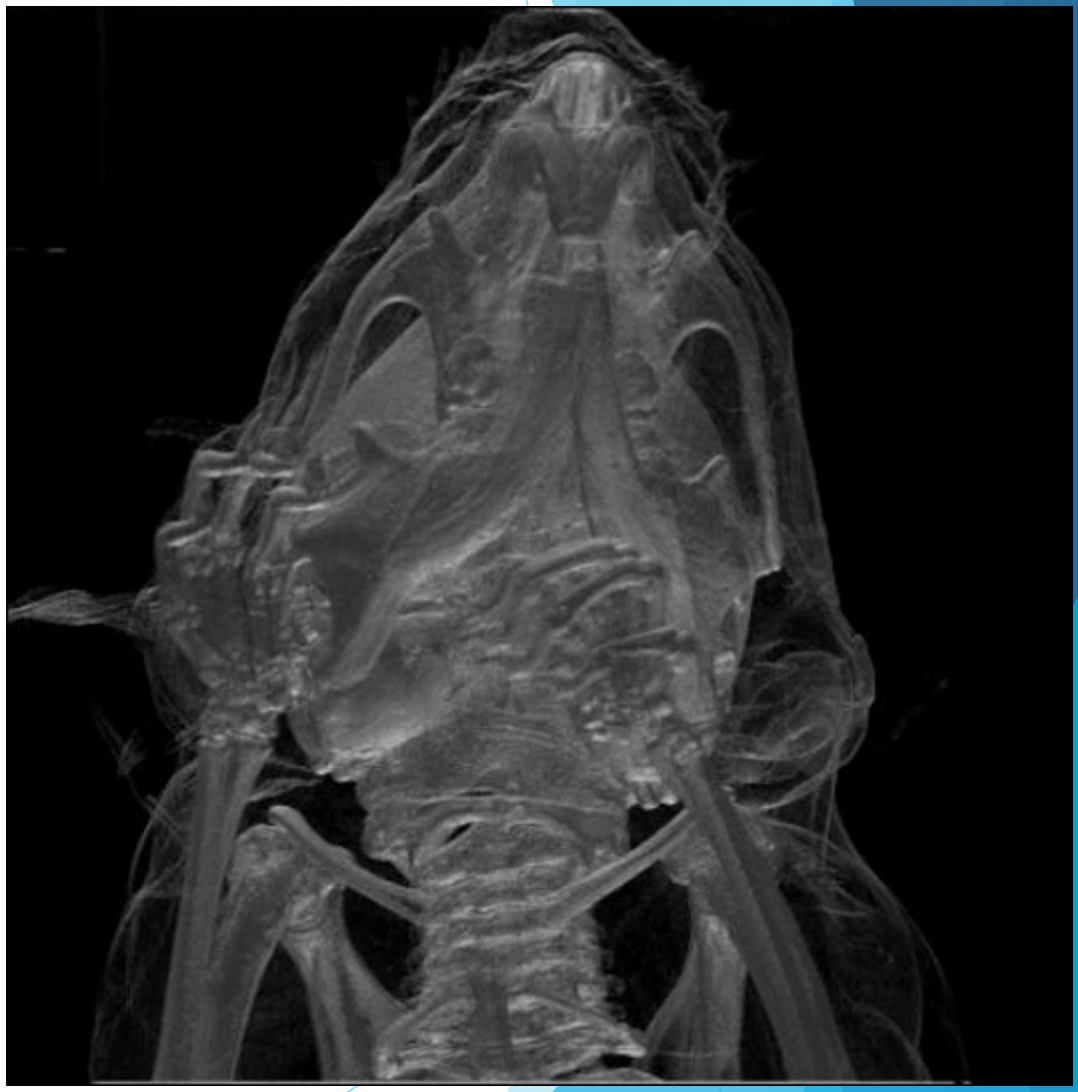
# Ray Tracing in ACG



# Volume Graphics

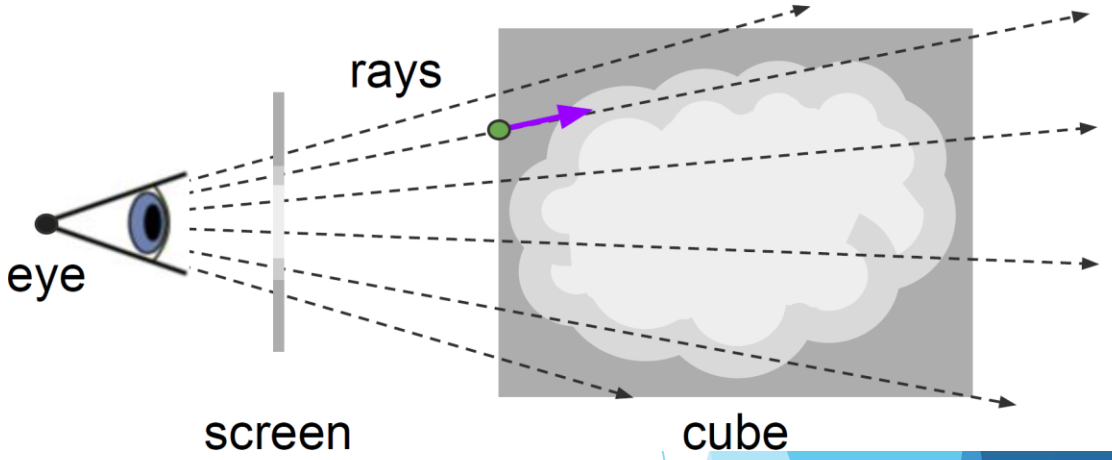
- Techniques for generating images from 3D scalar data
  - Originally motivated by scientific visualization
  - Data is acquired by measurement or generated by numerical simulation
  - Typical examples: computerized tomography (CT) or magnetic resonance imaging (MRI)
  - Other examples: fluids, gases, and natural phenomena like clouds, fog, and fire.

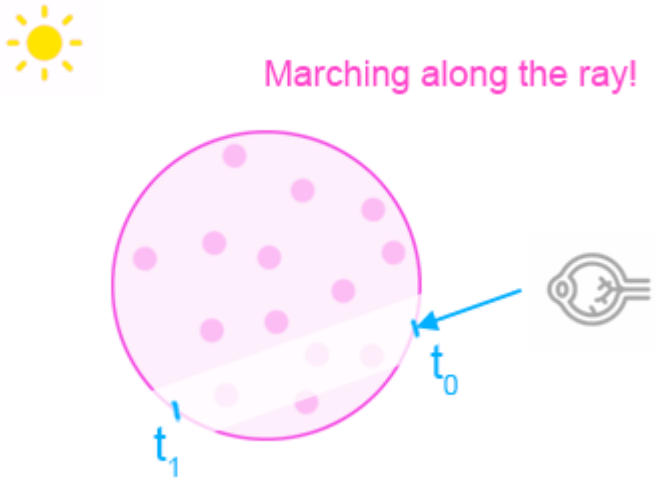


Animation from Wikipedia

# Volume Graphics: Basic Principle

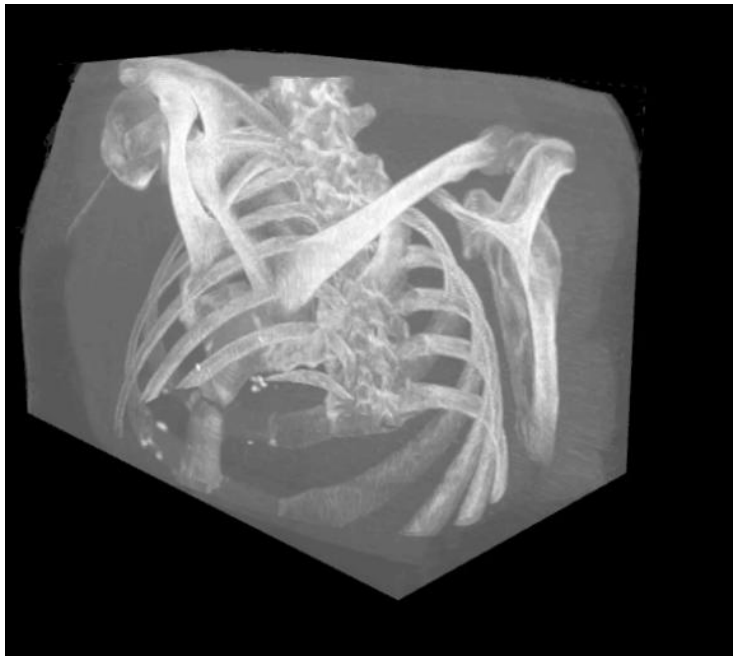
- Objective
  - Synthesize high-quality images from volumetric (i.e., 3D) data
- Several Volume Rendering algorithms
  - Photo-Realistic vs Non-Photo-Realistic (NPR)
  - CPU- or GPU-based
  - Based on Textures (slices) or on **Ray Casting**



# Volume Graphics Applications

# Medical Data Visualization



lightningchart.com

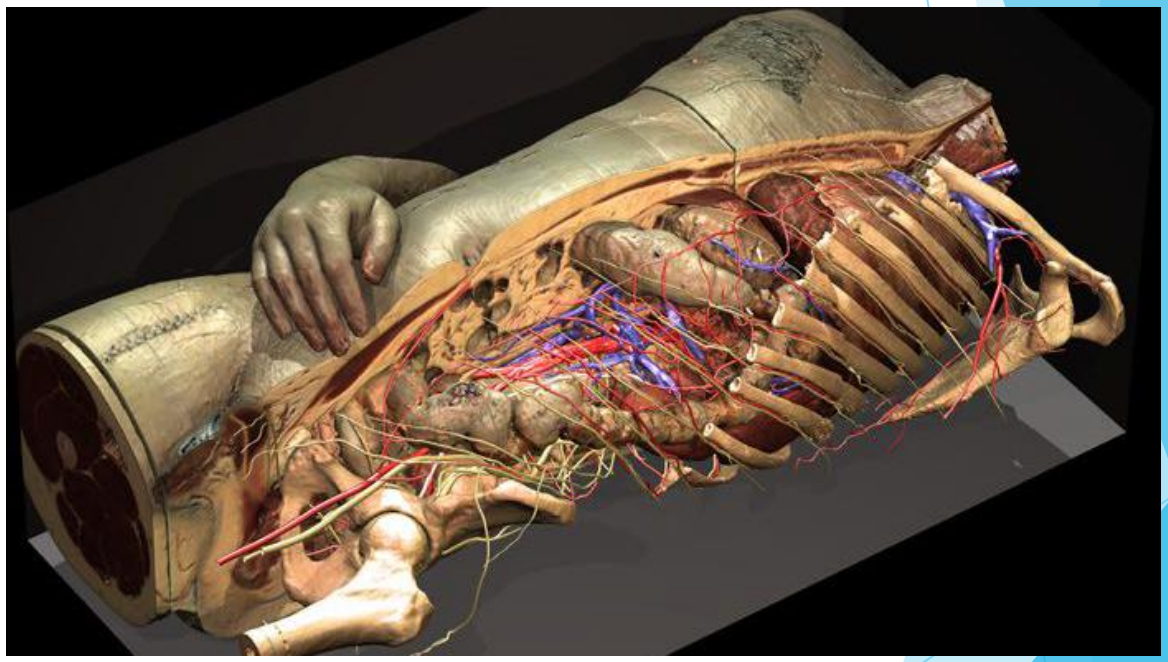


Image from Voxel-Man

# Clouds Rendering



Animation from Youngdo Lee's blog (Volume Rendeing)

# Volume Graphics in ACG



- Example result from the Volume Graphics project
- Includes
  - Real-time Volume Rendering (GPU)
  - Photo-realistic simulation of light propagation in participating media
  - Follows the laws of physics

# Conclusion

# Lecture's Summary

- Course Presentation
  - Team
  - Theory and Lab/Seminar content
  - Assessment and Calendar
- Course Content Overview
  - Basic Principles of Ray-Tracing
  - Basic Principles of Volume Graphics

# Key Take-Away Messages

- Demanding but very rewarding course
- Practical Approach to Learning
  - Oriented to the completion of 2 projects
  - Research-oriented component (project + presentation)
  - Progressive increasing autonomy (preparation for real-world professional requirements)
- **Make groups of 2!**

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# Credits

- Thank you to Josep Blat for making available the slides for ACG (2023)
- Thank you to Arnau Colom for providing the image on Project 1 (Ray-Tracing)
- Thank you to Víctor Ubieto for providing the image on Project 2 (Volume Graphics)



# The End

Questions?